

ACADEMIC TRACK: SOFTWARE ENGINEERING & CLOUD ARCHITECTURE

Event-Driven Microservices Architecture for High-Throughput Academic Assessment Platforms

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Abstract: This engineering design specification details an event-driven microservices architecture tailored for high-concurrency educational assessment platforms. By decoupling course submission pipelines from evaluation routing and cryptographic verification sidecars, the system ensures 99.95% uptime and sub-80ms API response times during submission deadlines. We present domain-driven decompositions, circuit breaker resilience patterns, and horizontal autoscaling configurations under simulated 50,000 concurrent user surges.

1. Domain Decompositions & Bounded Contexts

The platform architecture separates concerns into four bounded contexts: (1) Identity & Progressive Disclosure Governance; (2) Submission Ingestion & Storage; (3) Peer Review Assignment & Calibration Engine; and (4) Cryptographic Audit & Quorum Adjudication. Communication across bounded contexts utilizes asynchronous domain events published over Kafka message queues with idempotent consumer guarantees.

2. Failure Isolation & Resilience Engineering

To prevent cascading failures when external services experience transient latency spikes, each inter-service client employs Resilience4j circuit breakers with half-open probe states. If the cryptographic sidecar latency exceeds 250ms over a 20-call sliding window, requests degrade gracefully to asynchronous background queuing rather than rejecting client transactions.

Service Component	Runtime / Framework	Resilience Pattern	Target SLA	Avg p95 Latency	Autoscale Target
Gateway & Routing	Spring Cloud Gateway	Rate Limiter / Token Bucket	99.99%	12 ms	2-8 Replicas
Submission Core	Spring Boot 3 (Java 21)	Async Queue / Circuit Breaker	99.95%	48 ms	3-18 Replicas
Crypto Sidecar	PHP 8.2+ / OpenSSL	Isolated Process / Localhost	99.99%	18 ms	1:1 Co-located
Analytics Engine	Spring Data / JPA MySQL	Read-Replica Offloading	99.90%	65 ms	2-6 Replicas

3. Concurrency Benchmarks & Autoscaling Verification

Load testing was conducted using distributed k6 test harnesses simulating 50,000 students submitting 15 MB PDF reports within a 30-minute deadline window. The horizontal pod autoscaler (HPA) automatically scaled submission pods from 3 to 18 replicas based on 70% CPU threshold, sustaining p99 response times of 74ms with zero 5xx error spikes.

4. Architecture Decision Records (ADRs)

ADR-01 confirms the adoption of Client-Side Rendering (React+Vite) paired with Spring Boot REST to eliminate SSR multi-tenant cache leak vulnerabilities. ADR-02 establishes HMAC-SHA-256 tamper-evident ledgers for legal audit compliance.